Two Instruments, One Ecosystem

Open-source bioreactors: what they're *for*, and where OpenEvo goes wrong

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Biopunk Wetware Meeting ▪ `github.com/m9h/biopunk-bioreactor`

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The one-slide thesis

Chi.Bio and OpenEvo look like competitors. They aren't.

- **Chi.Bio is an instrument** — measures a culture richly, in situ, while holding it in a defined state.
- **OpenEvo is an actuator** — pushes a culture through selection, unattended, for weeks.

Both cost ∼\$300. Both are open. But a machine built for one purpose is *measurably worse* at the other.

The field serves three purposes, not one contest.

Three purposes

proto – Characterise

Hold a defined state; measure richly in situ; close feedback loops.

Champion: **Chi.Bio**

devo – Evolve

Weeks of unattended selection; produce *archived, replicated* evidence.

Champion: none yet

bior – Scale out

Many reliable reactors now; turnkey; supported.

Champions: Pioreactor, eVOLVER

proto → **devo** → **bior** — a lifecycle: prototype, evolve, produce.

Rich-&-few vs. cheap-&-many | non-contact vs. electrodes-OK | steady-state vs. run-to-stationary

Three lineages, one bench

Purpose	Founding lineage	Landmark demo
proto	chemostat as instrument + closed-loop control (Monod; Novick–Szilard; Milias-Argeitis '11)	optogenetic feedback holding expression + growth ('16)
devo	chemostat selection → LTEE → PACE (Novick–Szilard '50; Lenski '91; Esvelt '11)	aerobic citrate use in the LTEE ('12)
bior	continuous culture for production → arrays → eVOLVER (Herbert '56; Miller '13; Wong '18)	eVOLVER ePACE: compact Cas9 ('22)

Three traditions, seventy years — now one $300 device.

OpenEvo: credit where due

A real contribution — these are genuine strengths:

- Peak-to-peak growth readout tied to hardware dilution events (no local-maxima noise).
- 940 nm OD avoids optogenetic & ambient overlap; $R^2 \geq 0.98$ to $\mathrm{OD}_{600} = 6$.
- Closed autoclavable vessel — better sterility than Chi.Bio's open tube.
- 4,000-hour remote run; controlled from another continent.
- Assembly manual validated by an undergrad with no electronics experience.

The cheapest buildable three-media selection cycler in the field.
The problems below are correctable — framing and analysis, not hardware.

OpenEvo: two verifiable errors

Error 1 — the headline figure contradicts itself

Abstract: "55% faster." Results: "4.9 vs 7.6 h, a 36% increase in growth rate."
Same data. $7.6 \to 4.9\,\text{h}$ is a *36% shorter doubling time* **=** a *55% higher rate* ($1/4.9 \div 1/7.6 = 1.55$). The results sentence mislabels one as the other.

Error 2 — a false claim about a competitor

"Pioreactor... only the *software* is open source."
False: the hardware is openly licensed (CC BY-SA 4.0, per its `LICENSE`).
(Our own first draft misnamed it CERN-OHL — a contributor corrected it against the primary file.)

OpenEvo: the deeper gap — evidence

$n = 1$

One clone, one population sequenced. No replicates $\Rightarrow$ no convergence $\Rightarrow$ *cannot say which mutation caused the adaptation.* The LTEE runs **twelve** for this reason.

No archiving

One hand-taken glycerol stock. The LTEE's real innovation was the *freezer*, not the flask — revive an ancestor, race it against a descendant. Without it, fitness is unmeasurable.

OpenEvo cites Barrick's "... Archiving Populations, and Measuring Fitness" — then implements neither.

`devo` has no champion — yet

OpenEvo is the best *actuator*, not the champion: **devo is defined by evidence, not actuation.**

What the actuator must connect to (only one is new hardware)

- **replicate manifold** (borrow Pioreactor's cluster) → convergence
- **automated chilled archive** — *the one new module* → fossil record
- **sequencing** + **`breseq`** → correct variant calls
- **fitness loop** (reuses the OD hardware) → selection coefficients
- **selection design** (growth-coupling) → dodge the ALE trap

Champion = mostly connective tissue. Keep the actuator cheap.

The matrix (abridged)

	Chi.Bio	OpenEvo	Pioreactor	eVOLVER	EVE
Purpose	proto	devo	bior	bior	devo
Paper	yes	yes	none	yes	yes
Parallelism	8	1	cluster	16	3+
Rich sensing	spectro.	OD	plugin	config.	OD
Archiving	no	no	no	no	no
Cost	$300	$300	$329	<$2k/16	$115
Build	PCB	guided	buy	shop	mod.

Two patterns: nobody archives (the field's blind spot) · the only paperless platform is the only pure product.

The proposal: build two, share the parts

Characterisation machine

Non-contact is inviolable.

- + optode DO / pH (read through wall)
- + dispersive spectrometer 340–850 nm
- + sealed gas-controlled lid

Evolution machine

Cheap × many; resist sensing bloat.

- + 8 replicate chambers, one interface
- + **automated chilled archive** (nobody has built this)
- + batch-excursion mode; `breseq`

Why two? The sensors sort themselves by contact.

Optical (DO/pH/spectrometry) → characterisation. Electrode (conductivity/ERT/viable-cell) → evolution. *Share* the module footprint, sensor designs, firmware, calibration — not the vessel.

These are teaching instruments too

- **EVE** — eLife paper is literally "…and its educational use"; <\$200; validated in a *real classroom* evolving *E. coli* under chloramphenicol.
- **Pioreactor** — educators portal; one device teaches micro + EE + CS at once.
- **OpenEvo** — the manual *is* a course; anchored a global hackathon.
- **Chi.Bio** — "remove the lid, observe fluorescence by eye."

But the teaching inherits the research blind spots — archiving, replicates, and carrying capacity are as under-taught as they are under-built.

Where this meets CRISPR

- These tools **evolve** CRISPR — xCas9, base editors, compact Cas9, all PACE / eVOLVER (Hu '18; Thuronyi '19; Huang '22).
- CRISPR runs **inside** these tools — pooled screens in continuous culture; Cas-based growth-coupled selection.
- CRISPR may **close the archiving gap** — in-cell molecular recorders instead of the freezer.

The `bior` landmark is itself a CRISPR result — the intersection is already here.

The goal: self-driving evolution

OpenEvo *pitched* AI-directed evolution — "AI to supervise the evolution" — but built only the actuator.

The equivalence that makes it buildable

The observation layer an AI needs **is** the evidence apparatus that makes a `devo` champion.
Apparatus = champion; AI = autopilot. Build the apparatus first, then close the loop.

- **observe**: growth, fitness, genotype (archive → `breseq`) + in-cell CRISPR recorder
- **act**: media / light / dilution + **CRISPR-targeted mutation** (EvolvR — an IGI tool)
- **learn**: rule-based → Bayesian opt → RL, across the replicate fleet

Prove it: AI-directed vs. fixed schedule, same target, replicate fleet.
`docs/ai-crispr-evolution-project.md`

Take-aways / call to build

1. Stop comparing these as one contest — **three purposes**, different winners.
2. Evolution has no champion: nobody archives or replicates. *That's the opening.*
3. The killer feature nobody has shipped: an **automated chilled fraction collector**.
4. OpenEvo's errors are real but fixable — and we filed them, fairly, with fixes.

Public, CC BY 4.0, corrections welcome — a PR already landed and fixed one of *my* errors.

github.com/m9h/biopunk-bioreactor